Technical Case Study

Enterprise SaaS Architecture

Aircraft Appraisal Platform

Domain-specific SaaS solution automating aircraft valuation for certified appraisers, financial institutions, and aviation stakeholders

Aviation Technology

Enterprise Engineering

Document Classification

Technical Architecture

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Enterprise Aircraft Appraisal Platform

Domain-specific SaaS solution for automated aircraft valuation

◎ Core Value Proposition

- **Automated Valuation Engine** — Reduces appraisal time from **40+ hours to 2-4 hours**
- **Multi-Source Data Fusion** — Aggregates FAA + NTSB + Market Data + User Inputs
- **Reference Model System** — Reusable aircraft configuration templates
- **USPAP Report Generation** — Auto-generates compliance-ready DOCX reports
- **Collaborative Workflow** — Multi-user report sharing with RBAC

⚙ The Core Differentiator

The `AppraisalComputation` model implements a sophisticated **12-factor valuation algorithm** that calculates additions and deductions across airframe condition, engine residual value, propeller value, avionics, damage history, and more.

Platform Scale

150K+ Lines of Code	**80+** Database Tables
200+ API Endpoints	**80+** Domain Models

12-Factor Algorithm

- Airframe condition
- Propeller value
- De-ice systems
- Paint/interior
- Modifications
- AD compliance
- Engine residual
- Avionics equip.
- Damage history
- Maintenance
- Time-limited
- Add. equipment

Systemic Inefficiency in Aircraft Valuation

The macro-problems plaguing the aircraft appraisal industry

Fragmented Data Sources

Critical aircraft data exists across disparate systems requiring manual aggregation and cross-referencing, creating significant inefficiencies.

- FAA Registry (N-Numbers, ownership)
- NTSB Incident Database (damage history)
- Market Listings (comparable sales)
- Manufacturer Specs (baseline values)

Manual Calculation Overhead

Traditional appraisals require extensive manual computation across hundreds of valuation factors, introducing potential for human error.

40+	200+
Hours per Appraisal	Valuation Factors

Regulatory Compliance Burden

USPAP (Uniform Standards of Professional Appraisal Practice) compliance demands rigorous documented methodologies and comprehensive audit trails.

- ✓ Documented valuation methodologies
- ✓ Complete audit trails for all calculations
- ✓ USPAP-compliant report formatting

Market Volatility

Aircraft values fluctuate based on complex interdependent variables that must be continuously monitored and factored into valuations.

- Engine hours & TBO status
- Damage history impact
- Avionics upgrades & retrofits
- Real-time market conditions

Multi-Tier Architectural Blueprint

1 Presentation Layer

Vue.js 3 SPA
Report Builder, Lookup Tools, Real-time Validation

Blade Templates
Admin Dashboard, User Management

2 API Gateway Layer

Laravel Routes
200+ REST Endpoints, CSRF Protection

Middleware Stack
Auth, RBAC, ForceJsonResponse

3 Application Layer

- 13 Controllers
- 80+ Models
- Helpers

4 Persistence Layer

MySQL Database
80+ Tables, Complex Indexes, Eloquent ORM

File Storage
FAA CSV (75MB+), NTSB SQL, DOCX Reports

Data Lifecycle Journey

1. **User Action**
   Appraiser enters aircraft N-Number
2. **API Call**
   POST /lookup-aircraft endpoint
3. **Backend Processing**
   Query FAA, NTSB, reference models, market data
4. **Response**
   JSON with pre-populated form data
5. **Valuation Engine**
   Execute 12-factor calculation methods
6. **Report Generation**
   PhpWord generates USPAP-compliant DOCX

Orchestration

Vue.js State →
Laravel Session →
MySQL Store ✓

Inertia.js Props ▪ Eloquent ORM

The Stack: Technology Justification

Laravel
PHP 8.1

- Mature ecosystem with extensive package library
- Rapid development with artisan commands
- Strong ORM (Eloquent) for complex relationships
- Extensive middleware support for auth/RBAC

Use Case: Backend API, business logic, report generation

Vue.js 3
+ Inertia.js

- Modern SPA experience without full API complexity
- Reactive state management for 200+ form fields
- Real-time validation and computed values
- Seamless integration with Laravel backend

Use Case: Report builder wizard, lookup tools, dashboard

MySQL 8.0
Relational DB

- Relational integrity for complex aircraft data
- Complex joins across 80+ tables
- ACID compliance for financial calculations
- Proven scalability and backup strategies

Use Case: Persistent storage, transaction management

PhpWord
DOCX Generation

- Native PHP library for Word document creation
- USPAP-compliant formatting and styling
- Dynamic content injection from database
- Multi-page tables with proper pagination

Use Case: USPAP report generation, comparison charts

Blade + Vue
Hybrid Rendering

- Gradual migration path from legacy to modern
- Blade for admin, Vue for user-facing features
- Reduced risk during architecture transition
- Shared backend logic across both frontends

Use Case: Admin dashboard (Blade), Report builder (Vue)

Laravel Sanctum
Authentication

- Session-based authentication for SPA
- CSRF protection built-in
- Role-based access control (RBAC)
- Simpler than OAuth for single-tenant app

Use Case: User auth, report sharing permissions

Pragmatic Compromises & Trade-offs

Engineering decisions with clear rationale and impact assessment

MySQL over NoSQL

Relational vs. Document Store

Rationale: Complex relational data with 80+ tables requiring joins

Sacrifice: Schema flexibility for horizontal scaling

ACID compliance critical for financial calculations

Session-based Auth

over OAuth 2.0

Rationale: Simpler implementation for single-tenant application

Sacrifice: Third-party integration complexity

Laravel Sanctum provides sufficient security

Blade + Vue Hybrid

Gradual Migration Strategy

Rationale: Reduced risk during architecture transition

Sacrifice: Code duplication during transition period

Admin (Blade), User-facing (Vue)

Synchronous Report Gen

over Queue-based Async

Rationale: Immediate user feedback for report generation

Sacrifice: Scalability for large concurrent reports

Reports typically < 5 seconds to generate

Direct FAA Data Import

over External API

Rationale: No external API dependency or rate limiting

Sacrifice: Data freshness (monthly updates vs. real-time)

75MB+ CSV imports processed monthly

Engineering Philosophy

Every trade-off is **documented, measured, and reversible**. Decisions prioritize immediate business value while maintaining architectural flexibility for future evolution.

Evolution 1: Microservices Architecture

From monolithic Laravel to domain-driven microservices

Target Architecture

API Gateway (Kong/AWS APIG)

Request routing, rate limiting, authentication

Valuation Service (Go/Rust)

High-performance 12-factor calculation engine

Report Generation (Node.js/Python)

DOCX/PDF generation with template engine

Data Adapters

FAA Adapter, NTSB Adapter, Market Data Adapter

Benefits

Independent Scaling
Scale valuation service separately from report generation

Technology Diversity
Go for performance, Node.js for I/O, Python for ML

Fault Isolation
Service failures don't cascade to entire system

Current vs. Target

Architecture	Monolithic → Microservices
Language	PHP Only → Polyglot
Scaling	Vertical → Horizontal

Implementation Strategy

1. Extract valuation engine as standalone service
2. Create adapters for external data sources
3. Implement event-driven communication
4. Deploy with Kubernetes orchestration

Evolution 2 & 3: Real-Time & AI

Streaming data pipeline and ML-powered predictive valuation

Evolution 2: Real-Time Data Pipeline

Apache Kafka Streaming
Event streaming platform for real-time data ingestion

CDC for FAA Database
Change Data Capture for real-time registry updates

WebSocket Server
Push notifications for price changes and market events

Redis Caching Layer
Hot data caching for sub-millisecond lookups

Benefits: Real-time market value adjustments, instant notifications, live comparable sales data

Evolution 3: AI-Driven Predictive Valuation

Historical Sales Data
10+ years of training data

Feature Engineering
Airflow pipeline

TensorFlow Serving
Model deployment

A/B Testing
vs. rule-based

15-20%
Accuracy Improvement Potential

ML Model Features

- ✓ Automatic market trend adjustment
- ✓ Confidence intervals for value ranges
- ✓ Anomaly detection for unusual aircraft
- ✓ Seasonal price fluctuation prediction
- ✓ Regional market variance modeling

Implementation Timeline

Q1-Q2 — **Microservices Migration**
Extract valuation service, deploy K8s

Q3-Q4 — **Real-Time Pipeline**
Kafka, CDC, WebSocket implementation

Y2 — **AI/ML Integration**
Model training, TensorFlow Serving

Vision: Transform from reactive appraisal tool to predictive valuation platform

Platform Scale & Technology Stack

Key Metrics & Architecture Summary

150K+ Lines of Code	**80+** Database Tables	**200+** API Endpoints	**80+** Domain Models
12 Valuation Factors	**9** Aircraft Categories	**2** Report Templates	**4** Data Sources

Complete Technology Stack

- **Laravel 9** — PHP 8.1
- **Vue.js 3** — + Inertia.js
- **MySQL 8.0** — Relational DB
- **PhpWord** — DOCX Generation
- **Laravel Sanctum** — Authentication
- **Artisan** — Task Scheduling
- **Local Filesystem** — File Storage
- **Scribe** — API Documentation

Credit: Travel_Motion

Enterprise Aircraft Appraisal Platform

Transforming aircraft valuation through technical excellence